Skills Summary

The Rational Root Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — Which Numbers Are Even Possible

Theorem: every rational zero of a polynomial with integer coefficients has the form $\pm\frac{p}{q}$, where p divides the *constant term* and q divides the *leading coefficient*.

Build the list as: divisors of the constant, over divisors of the leading coefficient, each with both signs. When the leading coefficient is 1 the list is just the divisors of the constant.

Example: For $f(x) = 2x^3 + 3x^2 - 8x + 3$, list the candidates and test the simplest one.
Step 1. The theorem narrows an infinite search to a short list, so write the list first and only then start substituting.
Step 2. Divisors of 3 over divisors of 2 gives $\pm 1, \pm 3, \pm\frac{1}{2}, \pm\frac{3}{2}$. Testing $x = 1$: $2 + 3 - 8 + 3$.
$\Rightarrow$ $\boldsymbol{f(1) = 0}$, so 1 is a zero.

Try it: For $g(x) = 3x^3 - 4x^2 - 5x + 2$ the theorem allows $\pm 1, \pm 2, \pm\frac{1}{3}, \pm\frac{2}{3}$. Find $g(2)$.

check: $\boldsymbol{g(2) = 0}$

Skill 2 — Testing a Candidate Quickly

Two equivalent tests: substitute the candidate, or run synthetic division with it. Either way, a result of 0 means the candidate is a zero and $x - c$ is a factor; anything else means move on.

Synthetic division is worth the extra care: when it works, it hands you the quotient at the same time.

Example: Test $x = -1$ in $f(x) = x^3 - 2x^2 - 5x + 6$.
Step 1. Testing means evaluating; the only question is whether the result is zero, so compute f at the candidate and read the sign of the answer off the end.
Step 2. $(-1) - 2 + 5 + 6$. The result is not 0, so -1 is not a zero and $x + 1$ is not a factor.
$\Rightarrow$ **8**

Try it: Test $x = 2$ in the same $f(x) = x^3 - 2x^2 - 5x + 6$.

check: $\boldsymbol{-4}$

Skill 3 — Divide It Out and Finish

Once a zero c is found: divide by $x - c$. The quotient has degree one lower, and every remaining zero is a zero of that quotient.

A cubic drops to a quadratic, which you can factor or run through the quadratic formula — so you never need more than one lucky candidate.

Example: Factor $x^3 - 2x^2 - 5x + 6$ completely.
Step 1. The candidate list is the divisors of 6; $x = 1$ gives 0, so $x - 1$ is a factor and the job becomes a quadratic.
Step 2. Dividing by $x - 1$ leaves $x^2 - x - 6$, and that factors as $(x-3)(x+2)$.
$\Rightarrow$ $\boldsymbol{(x-1)(x-3)(x+2)}$

Try it: Factor $x^3 + 3x^2 - 4x - 12$ completely.

check: $\boldsymbol{(x-2)(x+2)(x+3)}$

Skill 4 — Reporting Every Zero

Count them: a degree-n polynomial has n zeros counted with multiplicity. If you have fewer, you are not finished.

And expect leftovers: the theorem finds only *rational* zeros. Irrational or non-real zeros never appear on the candidate list — they come out of the leftover quadratic.

Example: Find all zeros of $x^3 - 7x + 6$.
Step 1. Degree three means three zeros, so plan to find one from the candidate list and get the other two from the quadratic that is left.
Step 2. Divisors of 6: $x = 1$ gives $1 - 7 + 6 = 0$. Dividing by $x - 1$ leaves $x^2 + x - 6 = (x+3)(x-2)$.
$\Rightarrow$ $\boldsymbol{x = 1,\ 2,\ -3}$

Try it: Find all zeros of $2x^3 - 3x^2 - 11x + 6$.

check: $\boldsymbol{x = 3,\ \frac{1}{2},\ -2}$